

Linear Combination of Non-Unitaries

Theory, Computation and Applications

Session 2: Generalized constructions and Burgers' equation

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Structured matrix loading

An LCNU represents a matrix using terms that need not be unitary:

$$A = \sum_{\ell=1}^K c_{\ell} L_{\ell}.$$

- A direct Pauli expansion can require many terms.
- Sparse, structured matrices often admit shorter descriptions using special matrix classes (e.g. projectors and permutations arise in Finite Difference applications).
- Each term still needs an efficiently implementable unitary embedding.

Goal: Extend the LCNU construction of Session 1 (Kronecker products of simple projectors) to more general products of structured factors.

Notation and register order

$$\mathbb{P} = \{I, X, Y, Z\}, \quad \mathbb{S} = \{I, \rho_0, \rho_1, \rho_2, \rho_3\}.$$

Rho notation	Outer product	Sigma notation	Completion
ρ_0	$ 0\rangle\langle 0 $	$\sigma_+\sigma_-$	I
ρ_1	$ 0\rangle\langle 1 $	σ_+	X
ρ_2	$ 1\rangle\langle 0 $	σ_-	X
ρ_3	$ 1\rangle\langle 1 $	$\sigma_-\sigma_+$	I

The “sigma basis” was used in Session 1. We adopt ρ notation for more compact expressions.

Unitary completions and complements

Definition (Unitary Completion/Complement)

Suppose that $V \subset \mathbb{C}^N$ and $Q : V \rightarrow V$ is a linear operator such that $Q : W \rightarrow V$ is a unitary transformation into its image, where $W := \ker(Q)^\perp$. Then, a unitary operator $\overline{Q} : V \rightarrow V$ is said to be a unitary completion of Q if $\overline{Q}(w) = Q(w)$ for all $w \in W$. The unitary complement of Q is defined by $Q^\perp := \overline{Q} - Q$.

Comment. Such a Q is sometimes called a **partial isometry**. We will use this terminology when discussing products of operators.

The completion fills in the missing action on the kernel to obtain a unitary operator.

One ancilla gives a unitary block encoding

Let $R = LL^*$ and let $\bar{L}$ be a unitary completion. Then

$$U = \begin{pmatrix} L & L^\perp \\ L^\perp & L \end{pmatrix} = \underbrace{\begin{pmatrix} I - R & R \\ R & I - R \end{pmatrix}}_{U_1} \underbrace{\begin{pmatrix} 0 & \bar{L} \\ \bar{L} & 0 \end{pmatrix}}_{U_2}.$$

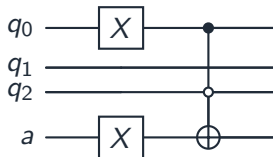
- $U_2 = X_a \otimes \bar{L}$: Depends on the choice of completion $\bar{L}$.
- $U_1 = I_a \otimes (I - R) + X_a \otimes R$: Depends only on L .

$$(\langle 0 |_a \otimes I) U (|0 \rangle_a \otimes I) = L.$$

The LCNU theory of Session 1 presented a systematic way to compute $\bar{L}$ and implement U for Kronecker products of sigma terms.

Recap: a sigma-product circuit

$$L = \rho_0 \otimes I \otimes \rho_2, \quad \bar{L} = I \otimes I \otimes X, \quad R = \rho_0 \otimes I \otimes \rho_3.$$



First gate column: U_2 . Second gate column: U_1 .

U_1 requires a control for each ρ term (ρ_0 open, ρ_3 closed).

Products and permutations

We want to retain the same completion and block-encoding construction while allowing

$$L = B_1 B_2 \cdots B_m, \quad B_i = \bigotimes_j P_{i,j}.$$

Each factor $P_{i,j}$ may be a Pauli matrix, a rho matrix ρ_r , or a permutation matrix on several qubits.

Every layer B_i acts on the **same n -qubit register**. Its factors may partition that register differently.

Existence of a completion and cost of its circuit are separate questions.

Completions of structured products

Theorem. For a product of the allowed factors,

$$\overline{L} = \overline{B}_1 \cdots \overline{B}_m, \quad \overline{B}_i = \bigotimes_j \overline{P}_{i,j}$$

is a unitary completion, where

$$\overline{P} = \begin{cases} I, & P \in \{\rho_0, \rho_3\}, \\ X, & P \in \{\rho_1, \rho_2\}, \\ P, & P \text{ is Pauli or a permutation.} \end{cases}$$

Easy extension. For unitary V, W , a completion of VLW is $V\overline{L}W$.

Why arbitrary intermediate unitaries fail

$$\rho_0 H \rho_0 = \frac{1}{\sqrt{2}} \rho_0, \quad \|\rho_0 H \rho_0 |0\rangle\| = \frac{1}{\sqrt{2}}.$$

This product is not an isometry on its nonzero initial space, so it has no unitary completion in the sense used here.

For partial isometries A, B , the product AB is a partial isometry exactly when

$$[A^* A, B B^*] = 0.$$

The structured matrices considered here satisfy this condition.

I. Erdelyi, “On Partial Isometries in Finite-Dimensional Euclidean Spaces,”

SIAM J. Appl. Math. **14**(3), 453–467 (1966).

Results on efficient implementation

Define the coordinate projectors

$$\mathcal{R}_n = \left\{ \bigotimes_{k=0}^{n-1} r_k : r_k \in \{\rho_0, \rho_3, I\} \right\}.$$

Theorem. If $LL^* = \sum_{j=1}^r R_j$, where r is a constant and $R_j \in \mathcal{R}_n$ and $R_j R_k = 0$ for $j \neq k$, then U_1 uses r multicontrol not gates.

LL^* factor	Circuit control
ρ_0	Open control
ρ_3	Closed control
I	No control

We want LL^* to be a sum of coordinate projectors. When does this happen?

A useful class of matrices

On n qubits, let

$$\mathcal{O}_n = \left\{ P_f \left(\bigotimes_{k=0}^{n-1} X^{b_k} \right) : f \in \text{Sym}(n), b_k \in \{0, 1\} \right\}.$$

These operations permute qubits and flip individual bits. They map a coordinate projector to another coordinate projector under conjugation.

$$O = P_{\text{swap}}(I \otimes X), \quad O(\rho_0 \otimes \rho_0)O^* = \rho_3 \otimes \rho_0.$$

Corollary. Products built from Pauli matrices, rho terms, and such permutations (also on subregisters) have $LL^* \in \mathcal{R}_n$.

U_1 needs at most one multi-controlled X for this special class.

Circuit construction

1. **Complete the factors.** Replace each ρ_r by I or X and preserve the product order to obtain $\bar{L}$.
2. **Find R .** Compute $R = LL^*$ and express it as disjoint coordinate projectors.
3. **Apply U_2 .** Implement $\bar{L}$ on the system and X on the ancilla.
4. **Apply U_1 .** Add a multicontrol NOT gate with a control for each ρ term (open or closed as needed).

Efficiency also requires an efficient circuit for $\bar{L}$. Arbitrary basis permutations can be expensive. Multi-controlled gates must also be synthesized into the chosen elementary gate set.

Four-qubit example: the completion

Let $S_2|x\rangle = |x + 1 \bmod 4\rangle$. Consider

$$L = (S_2 \otimes \rho_0 \otimes \rho_2)(\rho_0 \otimes \rho_0 \otimes I \otimes I).$$

Note that S_2 is a permutation (an incrementer). A little algebra shows that

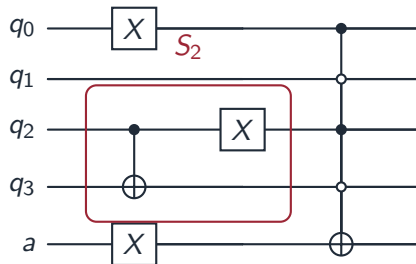
$$R = LL^* = \rho_0 \otimes \rho_3 \otimes \rho_0 \otimes \rho_3,$$

$$\bar{L} = S_2 \otimes I \otimes X.$$

U_2 and U_1 can be built using R and $\bar{L}$.

Four-qubit example: the full circuit

$$R = \rho_0 \otimes \rho_3 \otimes \rho_0 \otimes \rho_3, \quad \bar{L} = S_2 \otimes I \otimes X.$$



Burgers' equation and spatial discretization

On a periodic interval, with $N = 2^s$ grid points,

$$u_t = \nu u_{xx} - uu_x, \quad x_j = j\Delta x, \quad \Delta x = L_x/N.$$

Centered differences give, with indices modulo N ,

$$\dot{u}_j = \frac{\nu}{\Delta x^2}(u_{j+1} - 2u_j + u_{j-1}) - \frac{u_j}{2\Delta x}(u_{j+1} - u_{j-1}).$$

For $\mathbf{u} = (u_0, \dots, u_{N-1})^T$,

$$\dot{\mathbf{u}} = F_1 \mathbf{u} + F_2 \mathbf{u}^{\otimes 2}, \quad \mathbf{u}(0) = \mathbf{u}^0,$$

where $F_1 \in \mathbb{R}^{N \times N}$ and $F_2 \in \mathbb{R}^{N \times N^2}$.

Carleman linearization

Lifting to tensor powers produces a linear hierarchy:

$$\frac{d}{dt} \mathbf{u}^{\otimes j} = A_j^j \mathbf{u}^{\otimes j} + A_{j+1}^j \mathbf{u}^{\otimes(j+1)}.$$

$$A_j^j = \sum_{k=0}^{j-1} I_N^{\otimes k} \otimes F_1 \otimes I_N^{\otimes(j-k-1)},$$

$$A_{j+1}^j = \sum_{k=0}^{j-1} I_N^{\otimes k} \otimes F_2 \otimes I_N^{\otimes(j-k-1)}.$$

Truncate at order α by dropping the coupling to order $\alpha + 1$.

The finite linear system approximates the nonlinear dynamics.

R. Demirdjian, T. Hogancamp, D. Gunlycke,

An Efficient Decomposition of the Carleman Linearized Burgers' Equation, arXiv:2505.00285.

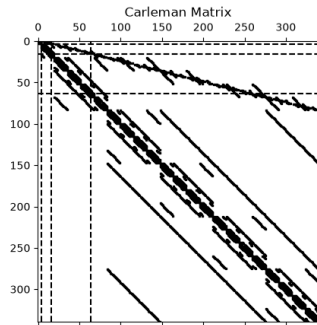
The truncated linear system

$$\dot{\mathbf{y}} = A\mathbf{y}, \quad \mathbf{y}(0) = (\mathbf{u}^0, (\mathbf{u}^0)^{\otimes 2}, \dots, (\mathbf{u}^0)^{\otimes \alpha})^T.$$

$$A = \begin{pmatrix} A_1^1 & A_2^1 & 0 & \cdots & 0 \\ 0 & A_2^2 & A_3^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & A_{\alpha-1}^{\alpha-1} & A_{\alpha}^{\alpha-1} \\ 0 & \cdots & \cdots & 0 & A_{\alpha}^{\alpha} \end{pmatrix}.$$

The j th block of $\mathbf{y}$ has N^j entries:

$$\Delta = \sum_{j=1}^{\alpha} N^j.$$



$$N = 4, \alpha = 4, \Delta = 340.$$

Structured blocks with nontrivial sparsity patterns.

Zero-padding to a qubit register

Set $D = 2N^\alpha = 2^{n'}$, $n' = 1 + \alpha s$, and

$$N_0 = D - \Delta = 2N^\alpha - \sum_{j=1}^{\alpha} N^j.$$

Then

$$A^{(e)} = \begin{pmatrix} 0_{N_0 \times N_0} & 0_{N_0 \times \Delta} \\ 0_{\Delta \times N_0} & A \end{pmatrix} \in \mathbb{R}^{D \times D}, \quad \mathbf{y}^{(e)} = \begin{pmatrix} 0 \\ \mathbf{y} \end{pmatrix}.$$

The padded coordinates remain zero when initialized to zero.

Example: $N = 4$, $\alpha = 2$ gives $\Delta = 20$, $D = 32$, $N_0 = 12$.

Backward Euler as one linear system

Use $M = 2^m$ stored time levels, including the initial state:

$$(I_D - \Delta t A^{(e)}) \mathbf{y}_k^{(e)} = \mathbf{y}_{k-1}^{(e)}, \quad k = 1, \dots, M-1.$$

$$\underbrace{\begin{pmatrix} I_D & 0 & \cdots & 0 \\ -I_D & I_D - \Delta t A^{(e)} & \ddots & \vdots \\ 0 & \ddots & \ddots & 0 \\ 0 & \cdots & -I_D & I_D - \Delta t A^{(e)} \end{pmatrix}}_{L^{(e)}} \underbrace{\begin{pmatrix} \mathbf{y}_0^{(e)} \\ \mathbf{y}_1^{(e)} \\ \vdots \\ \mathbf{y}_{M-1}^{(e)} \end{pmatrix}}_{\mathbf{Y}^{(e)}} = \begin{pmatrix} \mathbf{y}^{(e)}(0) \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

First level LCNU decomposition

The cyclic increment matrix

$$S_m = \begin{pmatrix} 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & 0 \\ 0 & \cdots & 0 & 1 & 0 \end{pmatrix}_{2^m \times 2^m}$$

$$L^{(e)} = L_1^{(e)} - \Delta t L_2^{(e)}$$

$$L_1^{(e)} = (I_M - S_m + \rho_1^{\otimes m}) \otimes I_D,$$

$$L_2^{(e)} = (I_M - \rho_0^{\otimes m}) \otimes A^{(e)}.$$

Note that ρ terms are used to conveniently remove unwanted (anti) diagonal entries.

Three LCNU terms for $L_1^{(e)}$, structure is independent of the time discretization.

Completions of the $L_1^{(e)}$ terms

c_ℓ	L_I	$\bar{L}_I$	$L_I L_I^*$
+1	$I_M \otimes I_D$	$I_M \otimes I_D$	$I_M \otimes I_D$
-1	$S_m \otimes I_D$	$S_m \otimes I_D$	$I_M \otimes I_D$
+1	$\rho_1^{\otimes m} \otimes I_D$	$X^{\otimes m} \otimes I_D$	$\rho_0^{\otimes m} \otimes I_D$

$$L_1^{(e)} = (I_M - S_m + \rho_1^{\otimes m}) \otimes I_D$$

Placing the first Carleman block

The first block is $A_1^1 = F_1$. Its contribution to $A^{(e)}$ is

$$\left(\rho_0 \otimes \rho_3^{\otimes(s-1)}\right)^{\otimes(\alpha-1)} \otimes \rho_3 \otimes F_1.$$

Example: $N = 4$, $\alpha = 2$. The register has five qubits and $N_0 = 12$.

$$\rho_0 \otimes \rho_3 \otimes \rho_3 \otimes F_1.$$

The projectors locate the block. The remaining factors implement its dynamics.
 F_1 can be further decomposed using two incrementers and an identity.

General construction: Demirdjian et al., Quantum Data Loading for Carleman Linearized Systems, arXiv:2605.00302.

Summary

- Explicit unitary completions for matrix and/or Kronecker products involving Pauli, rho, and Permutation matrices.
- Systematic rule for block-encoding LCNU decomposition using $\bar{L}$ and R .
- Efficiency implementations for structured classes of matrices.
- Our LCNU approach uses permutations and ρ matrices to encode time stepping and embedded Carleman blocks compactly.
- This method can treat general nonlinear dynamical systems (with polynomial nonlinearity).

A short decomposition is one ingredient of an efficient quantum solver.

Conditioning, coefficient normalization, state preparation, truncation error, and readout still determine the end-to-end cost. Future research efforts will investigate these important aspects.

References

1. I. Erdelyi. *On Partial Isometries in Finite-Dimensional Euclidean Spaces*. SIAM Journal on Applied Mathematics **14**(3), 453–467 (1966).
2. R. Demirdjian, T. Hogancamp, D. Gunlycke. *An Efficient Decomposition of the Carleman Linearized Burgers' Equation*. arXiv:2505.00285.
3. R. Demirdjian, T. Hogancamp, A. Gnanasekaran, A. Surana, D. Gunlycke. *Quantum Data Loading for Carleman Linearized Systems: Application to the Lattice-Boltzmann Equation*. arXiv:2605.00302.

LCNU Burgers' equation

Set $T = \rho_0 \otimes \rho_3^{\otimes(s-1)}$. The general decomposition reduces to

$$\begin{aligned}
 A^{(e)} = & \underbrace{\sum_{j=1}^{\alpha} T^{\otimes(\alpha-j)} \otimes \rho_3 \otimes A_j^j}_{\text{Linear contribution}} \\
 & + \underbrace{\sum_{j=1}^{\alpha-1} T^{\otimes(\alpha-j-1)} \otimes \rho_1 \otimes \left[(X^{\otimes s} \otimes I_{N^j}) A_{j+1}^{(e),j} \right]}_{\text{Quadratic contribution}},
 \end{aligned}$$

where

$$A_{j+1}^{(e),j} = \begin{pmatrix} A_{j+1}^j \\ 0_{(N^{j+1}-N^j) \times N^{j+1}} \end{pmatrix}, \quad \mathcal{P}_2 = X^{\otimes s}.$$

We use the general embedding from the lattice-Boltzmann paper to update the earlier Burgers construction.

Demirdjian et al., arXiv:2605.00302v4, Eqs. (26)–(27) and Appendix E.1.

Hands-on session

Solve Carleman linearized Burgers' equation using VQLS and LCNU

